EXPLORATORY DATA ANALYSIS ON A CARS DATASET

Minor Project Report

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Project	Exploratory Data Analysis on a Cars Dataset
Dataset	Car details v3.csv
Dataset source	https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-from-cardekho
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1. Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on a cars dataset. The work inspects the dataset structure, checks data quality, removes duplicate rows, handles missing values, summarizes important variables, visualizes distributions and relationships, and records evidence-based findings. A machine-learning prediction model is not required for this assignment.

Questions explored include: How are selling prices distributed? Which fuel types and transmission types are common? How do prices vary by transmission and model year? Is there a relationship between kilometres driven and selling price? Which numerical variables show stronger correlations?

2. Dataset Overview

The selected file is **Car details v3.csv**. The notebook output shows 8,128 rows and 13 columns before cleaning. The fields include name, year, selling_price, km_driven, fuel, seller_type, transmission, owner, mileage, engine, max_power, torque, and seats.

The dataset contains a mixture of categorical fields (for example, name, fuel, seller_type, transmission and owner) and numerical or numeric-like fields (for example, year, selling_price and km_driven). The vehicle specification fields mileage, engine, max_power and torque include text units and require conversion before numerical analysis.

Dataset source note: the CSV was loaded from the local Colab working directory. Add the original download page or provider URL here if it is available; the assignment asks students to clearly state the dataset source.

3. Data Quality and Cleaning

Initial checks used the dataset shape, column names, data types, missing-value counts, duplicate count, and descriptive statistics. The initial output showed 1,202 duplicate rows. Missing values were reported in mileage (221), engine (221), max_power (215), torque (222), and seats (221).

Cleaning steps performed: (1) copied the original DataFrame to preserve the raw data; (2) removed duplicate rows; (3) extracted numeric portions from mileage, engine and max_power; (4) extracted the first numeric portion

from torque for a basic numeric summary; and (5) filled missing values in mileage, engine, max_power, torque and seats with each column's median.

After cleaning, the notebook reported 6,926 rows and 13 columns, zero missing values across the displayed columns, and zero duplicate rows. The number of rows removed was 1,202. Torque values can use different units and conventions, so extracting only the first number is a simplification; torque should be interpreted cautiously in comparisons.

4. Descriptive Analysis

The numerical summary of the original data showed a mean model year of approximately 2013.8, a median year of 2015, and a mean selling price of approximately 638,272. The average values below are based on the cleaned dataset output captured in the notebook.

Average selling price by transmission

Transmission	Average selling price
Automatic	1,309,712.30
Manual	444,299.07

Average selling price by model year (selected years)

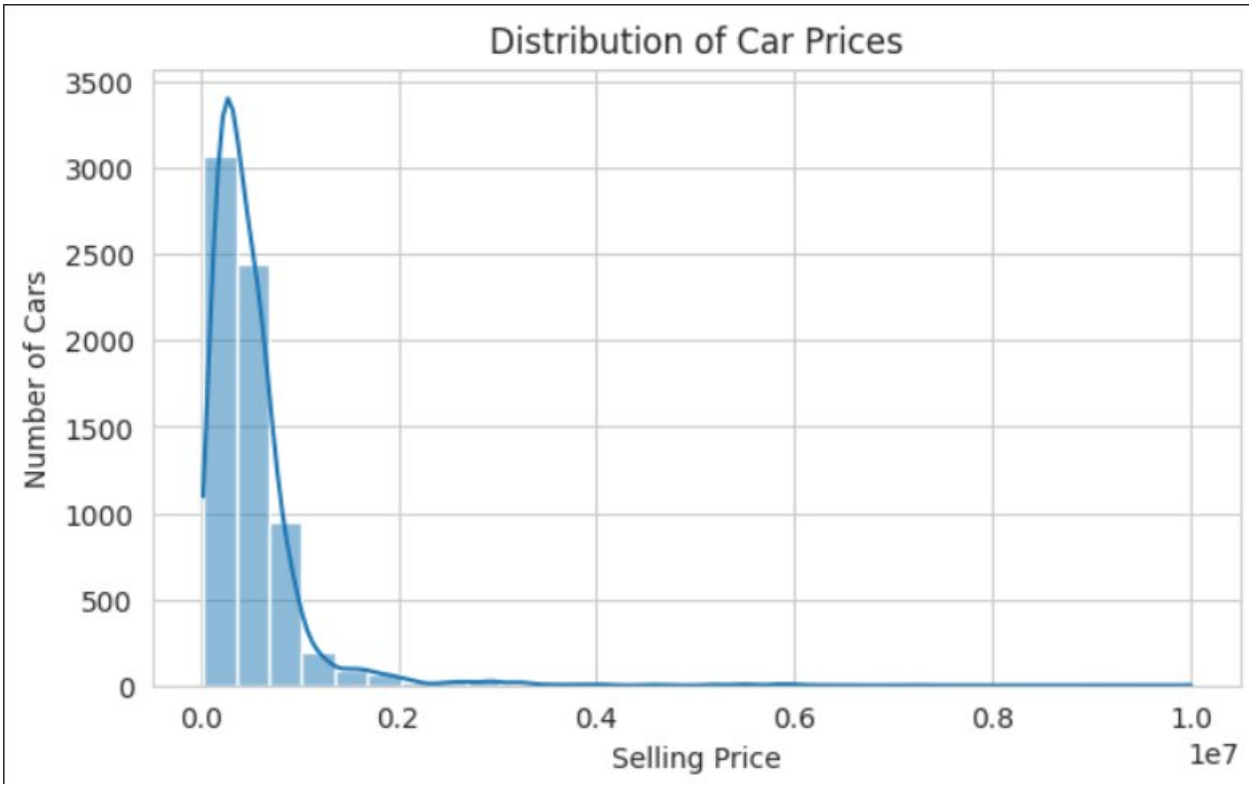
Year	Average price
2011	318,335.70
2012	351,387.38
2013	451,501.72
2014	511,858.80
2015	589,177.63
2016	653,180.34
2017	732,419.52
2018	866,376.61
2019	949,668.55
2020	888,349.16

The correlation between kilometres driven and selling price was reported as -0.166, which indicates a weak negative linear relationship in this dataset. Correlation does not establish causation and may be influenced by other variables such as model year, brand, fuel type, and vehicle condition.

5. Visual EDA

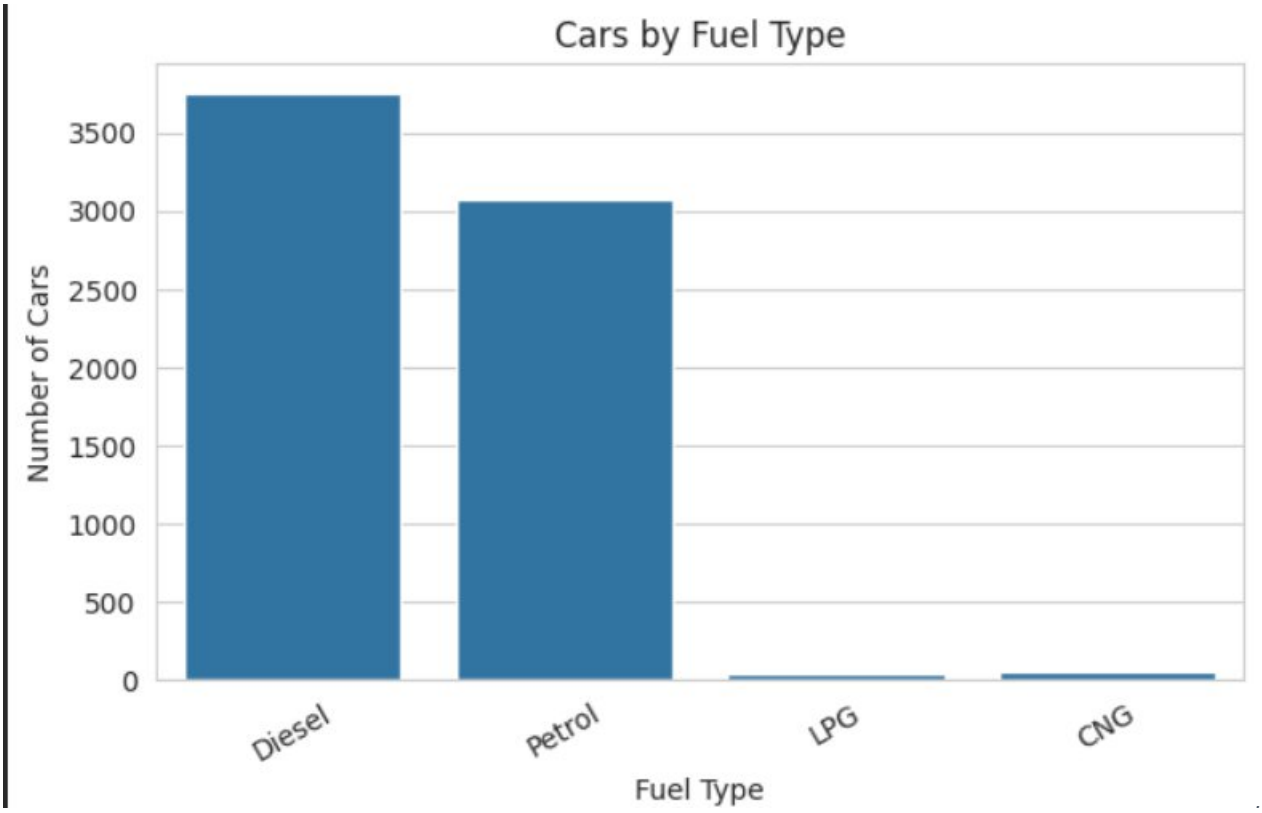
The notebook generated six charts. Insert the actual chart images from Colab in the spaces below, or copy the plots directly into this report before submission. The observations below are prompts and should be checked against the generated charts.

Chart 1. Distribution of Car Prices



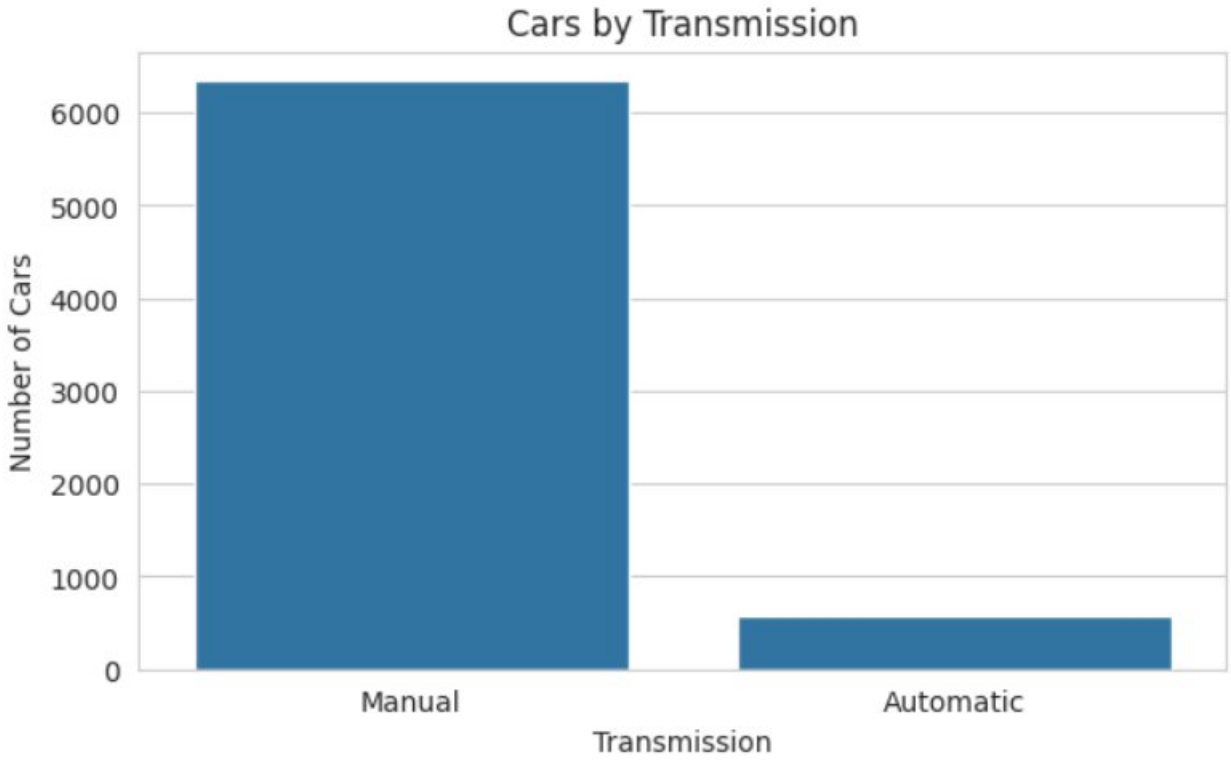
Observation: Check whether prices cluster toward the lower end and whether a long right tail or high-price outliers are visible.

Chart 2. Cars by Fuel Type



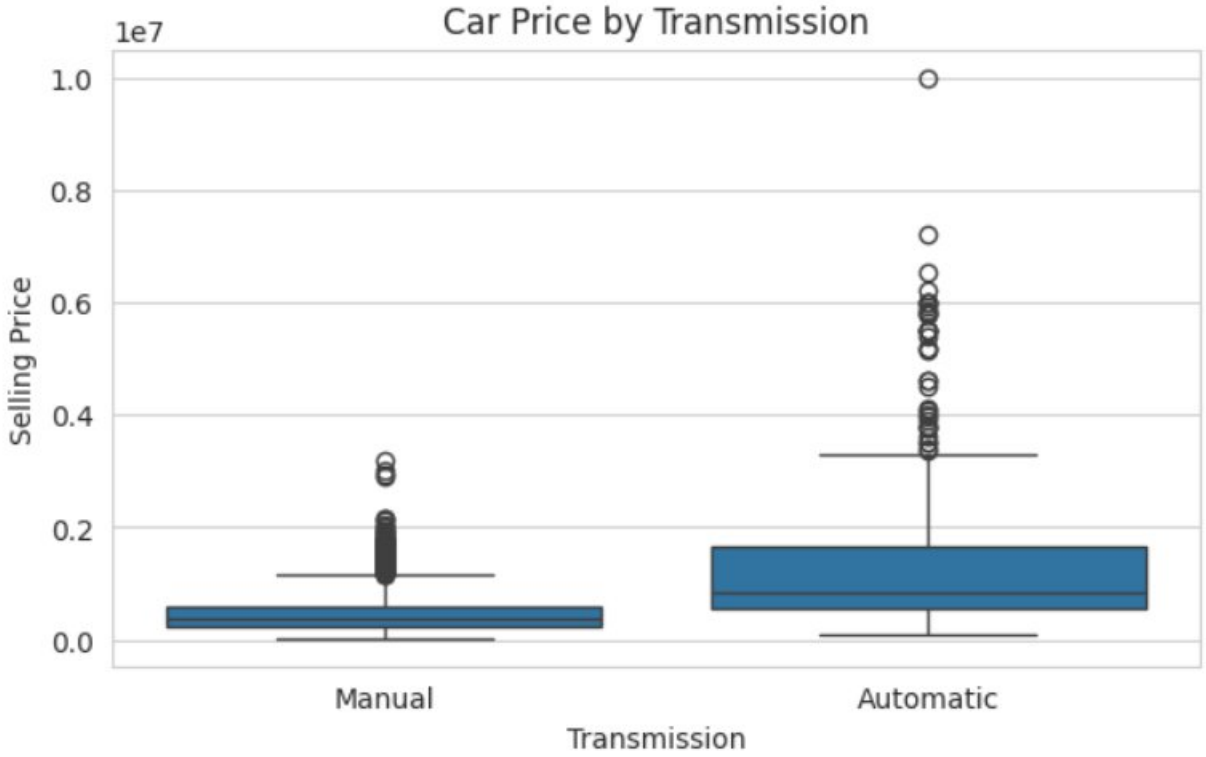
Observation: Identify the most frequent fuel category from the tallest bar.

Chart 3. Cars by Transmission



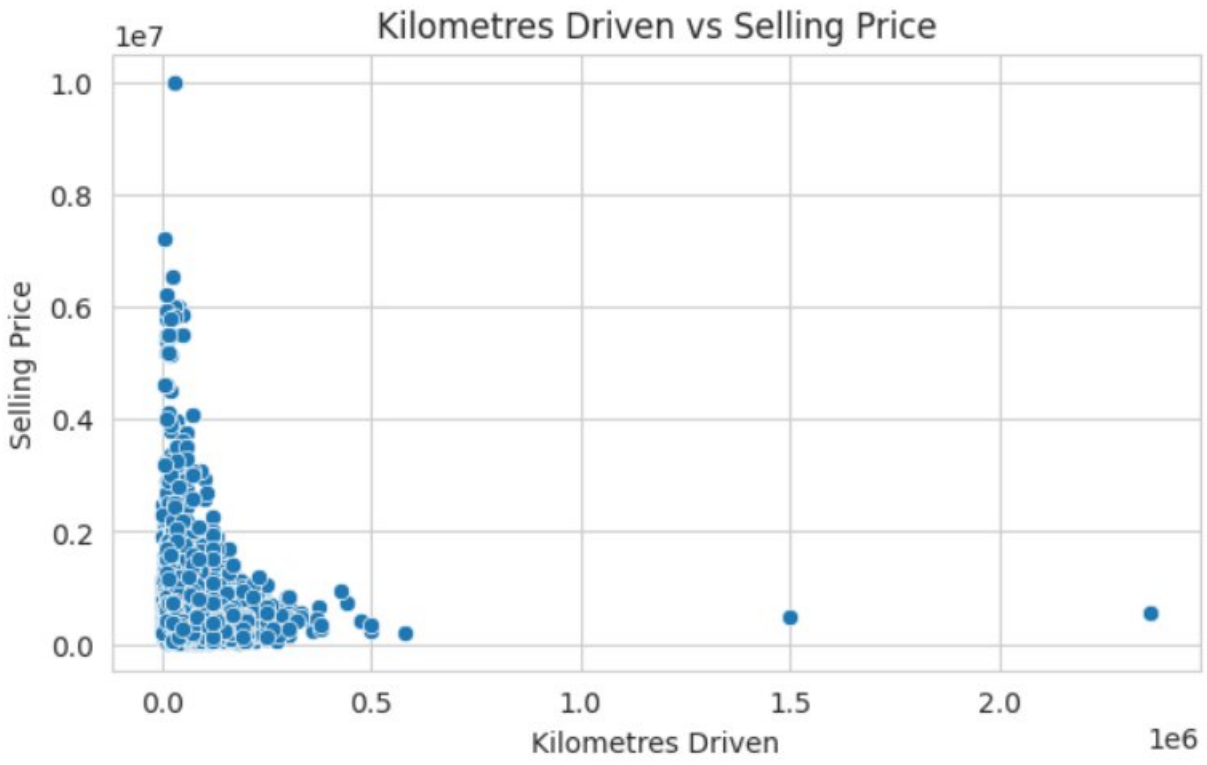
Observation: Compare the counts of automatic and manual vehicles.

Chart 4. Car Price by Transmission



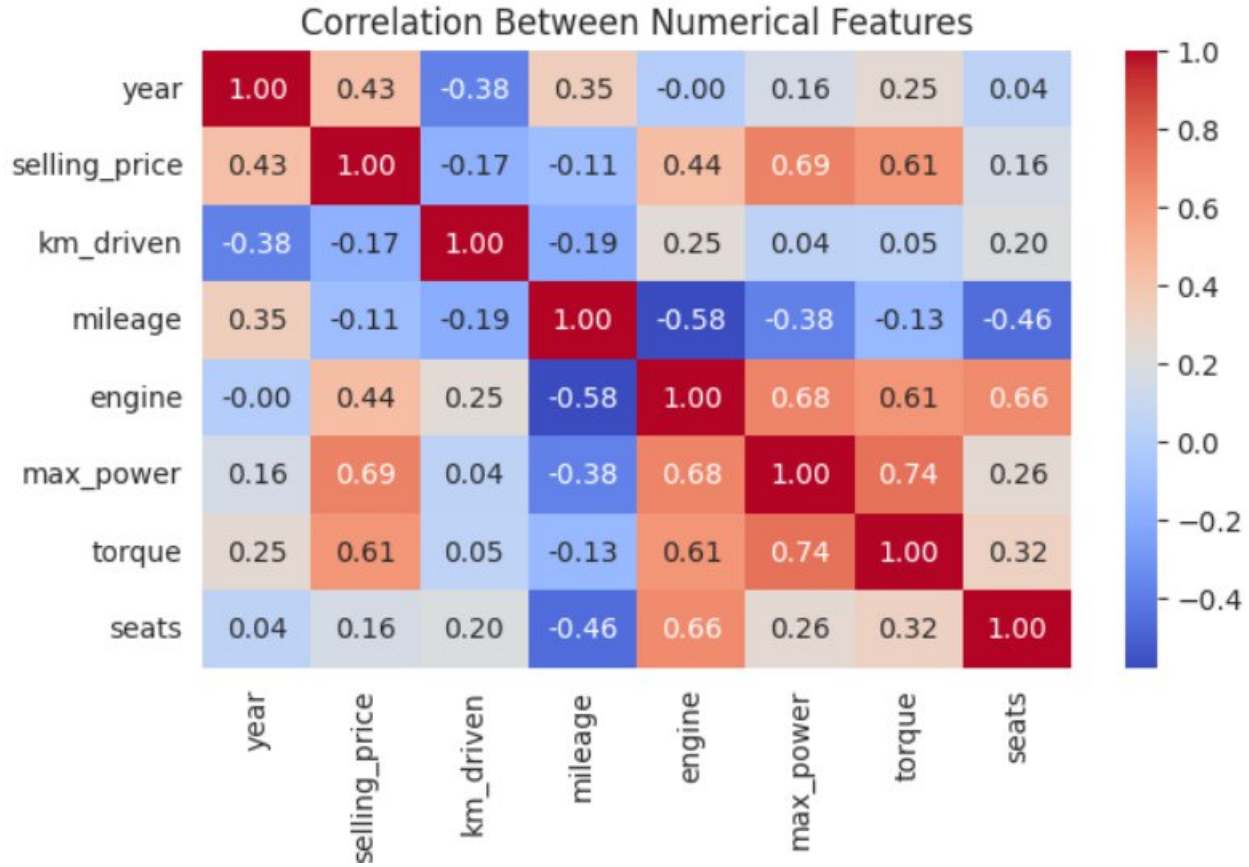
Observation: Compare the median, spread and outliers for automatic and manual cars; the box plot may show a higher price distribution for automatic cars.

Chart 5. Kilometres Driven vs Selling Price



Observation: The overall correlation is weakly negative (-0.166); look for substantial scatter and exceptions.

Chart 6. Correlation Between Numerical Features



Observation: Identify the strongest positive and negative correlation coefficients shown in the heatmap.

6. Key Findings

1. The original dataset contained 8,128 rows and 13 columns. Removing 1,202 duplicate records left 6,926 rows foranalysis.
2. Missing values were present in several vehicle-specification columns. Median imputation was applied to theselected numerical fields, and the final check reported zero missing values.
3. Automatic cars had an average selling price of 1,309,712.30, compared with 444,299.07 for manual cars. This is adescriptive difference in this dataset and does not prove that transmission alone determines price.
4. Average selling prices increased across the selected model years from 2011 through 2019, then declined in 2020.

Differences in the cars available for each year may affect these averages.

5. Kilometres driven and selling price had a correlation of -0.166, indicating a weak negative linear relationship.
6. Fuel-type frequency and numerical-feature relationships can be described using the generated count plot andcorrelation heatmap. Add the exact categories or coefficients visible in the plots before final submission.

7. Limitations

The analysis uses one dataset and may reflect its source, time period, sampling and reporting practices. Missing values were imputed with medians, which can reduce variability and may not represent the true values. Torque was simplified by extracting the first number from text strings; because units and formats may differ, torque-based comparisons would require additional standardization. Correlation does not imply causation. Extreme prices or kilometre readings should be investigated before deciding whether to remove them.

8. Conclusion

This project used EDA techniques to inspect and clean car data, calculate summary statistics, visualize distributions and relationships, and identify patterns in selling prices. Duplicate removal reduced the dataset from 8,128 to 6,926 rows, and the cleaning checks showed no remaining missing values or duplicate rows. The analysis found a weak negative relationship between kilometres driven and selling price and a large average-price difference between automatic and manual cars. Future work could standardize torque units, inspect outliers in context, compare average prices by brand and fuel type, and investigate multiple variables together.

References and Submission Notes

Dataset: Car details v3.csv. https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-from-cardekho